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PRIMARY RESEARCH PAPER

Temporal variation in macroinvertebrate assemblage structure in the Ohio River, USA

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Abstract A history of industrialization, oil spills, fires, and silt pollution from agriculture and urbanization led to the Ohio River being listed as one of the most polluted rivers in the USA. The Ohio River Valley Water Sanitation Commission (ORSANCO) and its members responded and conducted monitoring programs for pollutants to monitor water quality. ORSANCO collected annual macroinvertebrate samples at multiple locations using Hester–Dendy samplers. Collections using Hester–Dendy samplers from 1968 to 2004 resulted in 2,934,984 macroinvertebrates in 75 families and 321 species. We tested for temporal variation in trait (functional feeding groups) and taxonomic abundances of Ohio River macroinvertebrates. Our PCA analysis using trophic and tolerance traits resulted in significant temporal patterns. Macroinvertebrate richness increased following implementation of the federal Clean Water Act of 1972. Most trophic traits increased in abundance more recently. We interpreted these patterns as evidence for improvements in water quality and associated macroinvertebrate assemblages.

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Introduction

River pollution is a global environmental concern and is primarily fueled by industrial activities, urbanization, and agricultural practices. These human-induced sources along with natural factors contribute to the release of untreated or inadequately treated wastewater into river systems, significantly impacting water quality and ecosystem health (Halder & Islam, 2015; Al-Taai, 2021; Singh et al., 2022). Rivers can be contaminated with diverse pollutants, including heavy metals, nutrients, organic compounds, and microplastics. Heavy metals such as lead, chromium, nickel, manganese, and iron occur in river water at concentrations that exceed permissible limits (Al-Taai, 2021). Nutrient pollution, particularly from nitrogen and phosphorus, can cause eutrophication and algal blooms, disrupting aquatic ecosystems. River pollution has extensive consequences for aquatic ecosystems and biodiversity. The presence of contaminants can alter the composition of aquatic communities, including periphyton, macrophytes, macroinvertebrates, and fishes. Water contamination can contribute to variation in species richness and assemblage structure of most aquatic organisms (Sharp, 2018).

Long-term studies focused on stream macroinvertebrate communities are crucial for understanding temporal variation in ecosystems. These studies can provide in-depth insight into the complex dynamics of ecological responses, uncovering patterns and trends that may not be evident through short-term assessments (Jackson & Fureder, 2006). Monitoring variation in macroinvertebrate assemblages can help identify subtle shifts in assemblage structure and function that may act as early indicators of ecosystem stress (Resh et al., 1995). Usseglio-Polatera and Beisel (2002) found shifting macroinvertebrate assemblages, high biodiversity, and good water quality in the upper reaches of the Meuse River (France, Belgium, and Netherlands) in a long-term study. However, human impacts in some areas led to habitat decline. Vaughan & Ormerod (2012) found increased family-level richness from 1991 to 2008 in British river macroinvertebrates, and attributed variation to improved water quality and hydrology. Although human impacts result in negative impacts on macroinvertebrate assemblages, reductions in water pollution are predicted to positively impact macroinvertebrate assemblages. Our interest was a similar study of macroinvertebrate assemblages in the Ohio River, USA. We assessed temporal variation in the macroinvertebrate assemblage of the Ohio River following the enactment of the Clean Water Act in 1972. Our hypotheses were to (1) test for temporal variation in taxon abundance and taxon richness with the expectation that richness peaked after implementing the Clean Water Act; and (2) test for temporal variation in abundances by taxon and by trophic traits of macroinvertebrate assemblages with expectations that increased abundances are an indicator of improved water quality.

Material and methods

Study area

The Ohio River is one of the largest rivers in the USA at 1,579 km long (Dailey et al., 2014). Mean depth is 7.3 m, the deepest point is 40 m, and the watershed is 528,000 km^2 (Gupta et al., 2005). The river is formed by the confluence of the Allegheny and Monongahela Rivers in Pittsburgh, Pennsylvania. It flows southwest along the borders of Pennsylvania, West Virginia, Kentucky, Ohio, Indiana, and Illinois to Cairo, Illinois (Fig. 1). Many anthropogenic elements contribute to complicated environmental issues that demand management and regulation. Over 30 million people, about 10% of the US population, reside in the Ohio

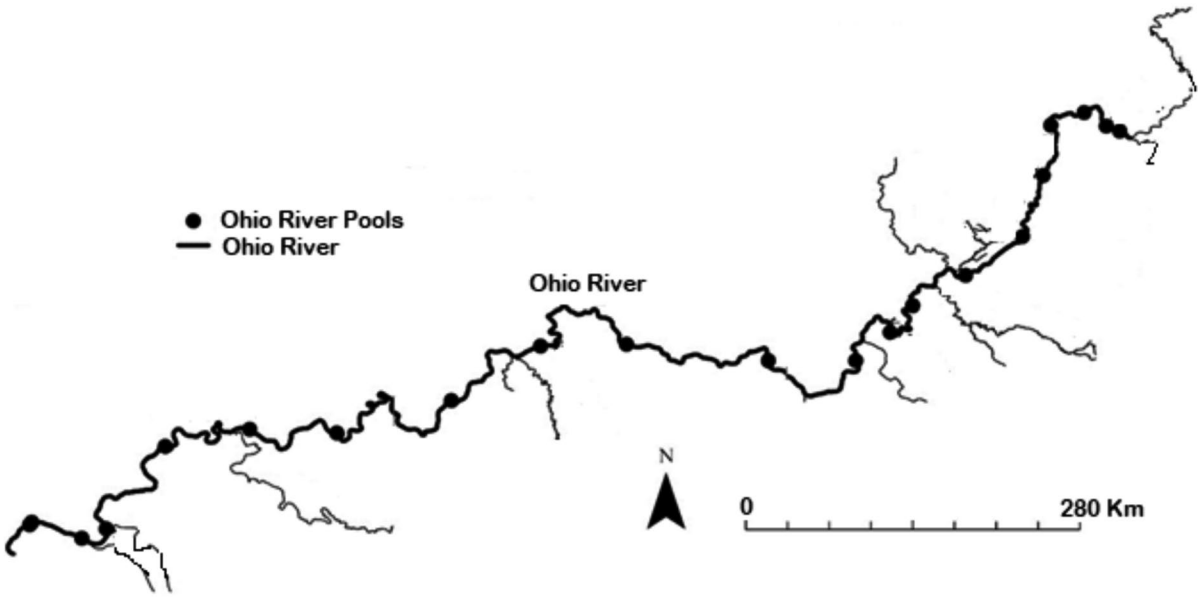

Fig. 1 Sites on the Ohio River where ORSANCO collected macroinvertebrates

River Basin (Applegate et al., 2007)., and the river supplies drinking water for more than 5 million people (ORSANCO, 2018). Twenty navigational dams allow barges to transport millions of tons of cargo, including coal. Almost 600 licensed dischargers occur in the Ohio River basin, representing a wide range of sectors, including power plants, municipal wastewater treatment plants, and manufacturers of metals and chemicals. For example, there are 49 power production sites on the river, 194 municipal wastewater treatment facilities, 383 industrial discharges, and 114 contaminated process outflows (Wooten et al., 2006). Further, abandoned mines, agricultural practices, and urban runoff are the main contributors to the Ohio River water pollution (ORSANCO, 2018).

The Ohio River Basin was drastically modified with logging and draining of wetlands following European colonization (Thomas et al., 2005). Land use was dominated by agriculture, but agriculture was converted to forest during the 1960s–80 s (Tayyebi et al., 2015). Urban and agricultural land use in the Ohio River Basin were 13% and 74%, respectively, in 1930, exceeding critical threshold effects on water quality and aquatic health (Tayyebi et al., 2015). Thomas et al. (2005) found improvements in the fish assemblages of the Ohio River in the most recent 50 years.

Macroinvertebrate collection and sampling procedure

The Ohio River Valley Water Sanitation Commission (ORSANCO) collected macroinvertebrates as part of its monitoring activities to identify contaminants that can obstruct specific river uses (ORSANCO, 2023). Applegate et al. (2007) summarized details for sampling locations and collections. ORSANCO used a random probability-based strategy to choose 15 sampling locations in each navigational pool. Samplers were placed at locations downstream of wastewater outfalls (Applegate et al., 2007). Data were collected yearly at two to three sites of the 20 navigational pools (ORSANCO, 2023). Macroinvertebrates were sampled using multiple sampling methods including Hester Dendy (HD). We focused on macroinvertebrates collected using the HD sampling method based on the longest duration, and other collection methods were not comparable. The HD samplers were constructed from tempered masonite cardboard cut into 7.6 cm^2 plates and 2.5 cm square spacers (ORSANCO, 2023). Pavers with five HDs were positioned on the riverbank at 3 m depth and secured to the shore at the downstream end of 500 m sampling locations. Samples were submerged in the river for six weeks in late August or early September. Samplers were retrieved and specimens scraped into a bucket of water (ORSANCO, 2023). A standard #30 sieve was used to filtered specimens into 70% ethanol for later identification and enumeration. ORSANCO makes publicly available macroinvertebrate data in the Ohio River (https://www.orsanco.org/data/macroinvertebrates/).

Data analysis

We used a family-level approach to quantify temporal variation in taxa because all taxa were not identified to genus-level. First we tested for correlations between taxon family richness and collection year. Then, we summarized and visualized taxonomic composition using principal components analysis (PCA), followed by Pearson correlations of PCA axes with collection year (Artz at el., 2020). We used Minitab 20.3 (minitab.com) for all analyses.

We categorized macroinvertebrate families based on their trophic traits and analyzed how these traits varied temporally. We used PCA followed by correlations of PCA axes with collection year, and we tested for correlations of trophic trait abundances with collection year. The trophic traits were from Merritt et al. (2017) as collectors–gatherers, filterers, predators, scrapers, and shredders. We did not apply the commonly used 5% rarity threshold for taxa (Campbell et al., 2011; Shade and Gilbert, 2015) as it would reduce the assemblage to only six families. Although zebra mussels [*Dreissena polymorpha* (Pallas, 1771)] are the most abundant macroinvertebrate taxon in the Ohio River and can negatively affect other invertebrates by altering habitat complexity and interspecific interactions (Gonzalez and Downing, 2007; Kolar et al. 2002), we excluded data on zebra mussels from our analysis to focus on the temporal changes non-invasive species (Jackson et al. 2010). Further, their inclusion in our analysis might prevent our ability to discover subtle shifts in the assemblages over time.

Results

Taxon richness

A total of 321 macroinvertebrate species were represented by 2,934,984 individuals in 9 phyla, 14 classes, 29 orders, 75 families, and 173 genera. The Chironomidae family had the highest relative abundance at 22%. Naididae and Gammaridae relative abundances were 4%. Taxon richness was significantly correlated with collection year (r = 0.829, $P < 0.001$; Fig. 2).

The ordination using taxonomy resulted in a first PC axis explaining 50% of variation and the second PC axis explaining 8.8% of variation (Fig. 3). Scores on the first PC axis were significantly correlated with year (r = 0.75, $P < 0.001$). Macroinvertebrate taxa with higher abundance in recent years were Caenidae, Siadidae, Coenagrionidae, and Chironomidae. Macroinvertebrates with higher counts in earlier years were Simuliidae.

We found that all traits of macroinvertebrates increased in abundance from 1968 to 2003 (Fig. 4). All trophic traits were significantly correlated with year (Collector filterers r = 0.060, $P < 0.001$; Collector gatherers r = 0.70, $P < 0.001$; Predators r = 0.57, $P < 0.001$; Scrapers r = 0.88, $P < 0.001$; Shredders r = 0.83, $P < 0.001$). Collector filterers were the most abundant trophic trait (64%). Collector gatherers ranked second (14%), and shredders, scrapers, and predators followed with 14%, 7% and 1% of the total population, respectively. The abundance of collector filterers in the Ohio River was low from 1967 to 1970 and decreased in 1977. However, from 1971 to 1976, there was an increase in the abundance of collector filterers, with the highest abundance from 1997 to 1999. The abundance of collector gatherers increased from 1979 to 2002, except for a slight decline in 1968 and 1977. The abundance of predators was low in 1975, 1976, and 1980. Predator abundance increased between 1982 and 1991 and varied until 2002. Abundance of scrapers decreased from 1970 to 1973 and fluctuated until 2002. Shredders abundance decreased between 1968 and 1969.

The PCA using trophic traits resulted in a first PC axis that explained 84.3% of variation, and a second PC axis that explained 8.8% of variation (Fig. 5). The first PC axis was significantly correlated with year (r = 0.78, $P < 0.001$). Collector-filterers, collector-gatherers, predators, scrapers, and shredders loaded positively on both PC axes (Fig. 5).

Discussion

We observed significant temporal changes in the macroinvertebrate assemblage structure of the Ohio River during a 35-year period. Increased taxon richness and trait richness in recent years suggests improvement

Fig. 2 Correlation analysis of family macroinvertebrate richness for all sites with collection year (r = 0.829, $P < 0.001$)

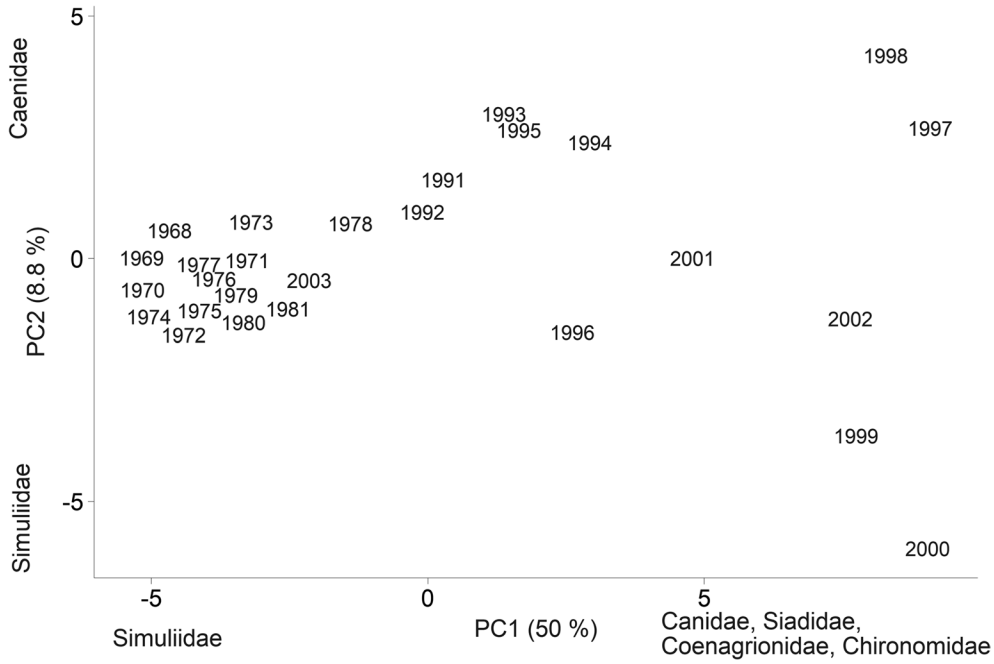

Fig. 3 Principal components analysis of Ohio River macroinvertebrate assemblages based on taxonomy from 1968 to 2003

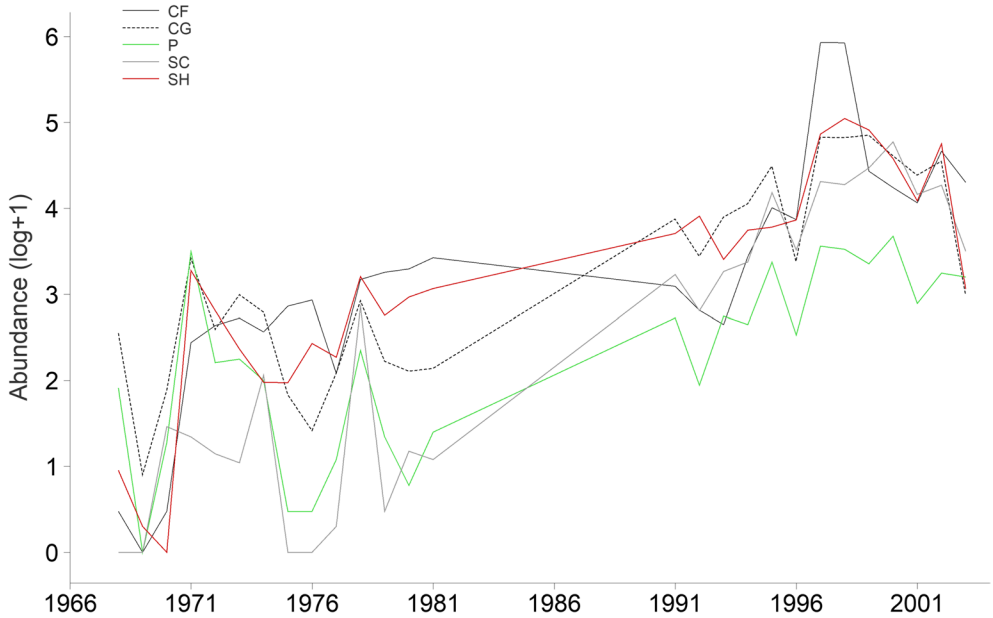

Fig. 4 Log-transformed abundance of macroinvertebrates classified into trophic trait categories. Trophic traits are collector filterers (CF), collector gatherers (CG), predators (P), scrapers (SC), and shredders (SH)

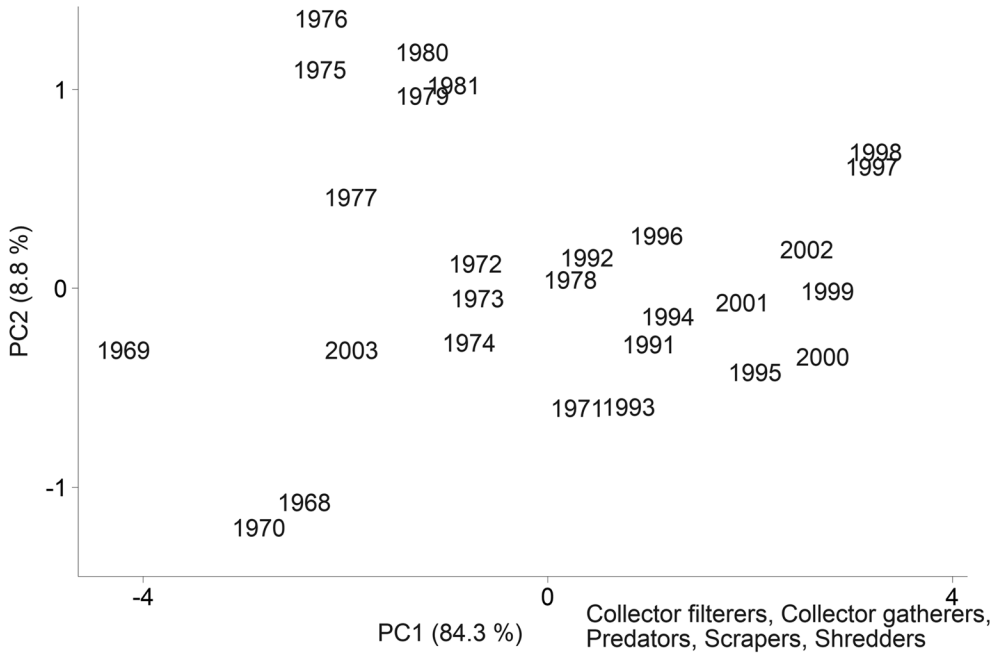

Fig. 5 Principal components analysis of Ohio River macroinvertebrate assemblages based on traits from 1968 to 2003

of the Ohio River ecosystem over historic conditions (McCord & Kuhl, 2013). We found significant variation in abundances of several macroinvertebrate traits. These results corroborate previous research that highlighted the importance of taxonomic richness and trophic traits of benthic organisms in evaluating the health of aquatic ecosystems (Pallottini et al., 2017; Archaimbault et al., 2010; Mondy et al., 2012). Our study provides support for environmental regulations such as the Clean Water Act of 1972.

Increased taxon richness of the Ohio River macroinvertebrate assemblages provides potential for increased understanding of ecological processes. Increased taxon richness likely had far-reaching impacts on the river ecosystem (Dodds et al., 2008). We observed an upward trend in taxon richness that began in the mid-1970s, coinciding with the implementation of the Clean Water Act (CWA). The CWA required a reduction of pollution and restoration of habitats, which likely contributed to the observed increase in taxon richness (Allan, 2004).

We found increased abundances of Chironomidae in recent years. Other studies found increased recent abundances of Chironomidae in other rivers, with ecological dominance of ecosystems (Smith et al., 2023). Several Chironomidae taxa were observed to increase in abundance during this time period: *Gammarus fasciatus* Say, 1818 (Gammaride) abundances increase with habitat suitability or improved environmental conditions (Edwards et al., 2024). *Glyptotendipe*s spp. (Chironomidae) and *Dicrotendipes nervosus* (Staeger, 1839) (Chironomidae) respond to environmental changes or emerging ecological opportunities (Miller et al., 2021; Robinson, 2020).

Recent studies highlighted the crucial role of sensitive taxa such as collector macroinvertebrates in diagnosing water quality (Tiemo et al., 2019; Ramírez & Gutiérrez-Fonseca, 2014; Prashant, 2020). Collectors actively participate in maintaining water clarity and nutrient cycling, significantly impacting ecosystem stability and resilience. Collector gatherers were less prevalent than collector filters in the Ohio River, yet are vital in processing organic matter and cycling nutrients in aquatic habitats (Bellamy et al., 2019). Involvement of filter feeders in detritus processing and nutrient release underscores their significance in sustaining ecosystem productivity and function (Thrush et al., 2006). Filter feeders efficiently remove suspended particles and organic matter from aquatic environments, enhancing water quality and promoting nutrient availability for other organisms, shaping ecosystem

dynamics (Yoon et al., 2022; Brown et al., 2023). Filter feeders have ecological importance in supporting microbial communities and facilitating nutrient recycling processes (Johnson et al., 2020; Brown et al., 2023). Scrapers, shredders, and predators represented low abundance trophic traits among aquatic macroinvertebrates of the Ohio River. Scrapers contribute to algae control and substrate stability by grazing algae and preventing excessive growth. Shredders enhance detritus breakdown and nutrient cycling by fragmenting organic matter into smaller pieces, accelerating decomposition processes. Predators regulate prey populations, exerting top-down control on assemblage structure, influencing trophic interactions and biodiversity. Their presence may maintain prey populations at sustainable levels and prevent single species from dominating the ecosystem (Garcia et al., 2022). Scrapers are indicators of sites with higher water quality, while extremely high abundance of predators indicates poorer water quality due to anthropogenic activities (Jimenez et al., 2021). Increased abundance of collector filterers, scrapers, and shredders in recent years implies that each of these groups contributes a unique role to the ecosystem. These roles may contribute differently to ecosystem functions such as nutrient cycling, energy flow, or habitat structure. Understanding these distinctions is crucial for comprehending the dynamics and stability of the ecosystem.

Although we found increased taxon richness for macroinvertebrates in the Ohio River from 1968 to 2003 after implementation of the Clean Water Act, specific human impacts on macroinvertebrates were not obvious. Angradi et al. (2010) were unable to identify macroinvertebrate assemblage responses to urbanization for three large rivers. However, Angradi et al. (2010) identified several invertebrates intolerant to pollution that increased in abundance in urbanized reaches. Angradi et al. (2010) suggested that urbanization effects included nutrient enrichment and an impervious surface threshold near 5% where urban-intolerant taxa were rare and urban tolerant taxa were abundant. Urban zones in the Ohio River were more frequent and closer together than on the Mississippi and Missouri rivers. We conclude that macroinvertebrate assemblages of the Ohio River improved from 1968 to 2003, yet there are obvious human impacts that limit the river from fully recovering.

Conflict of interests

The authors declare that they have no competing financial interest or personal relationship that could influence this manuscript.

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Data availability Data are publicly available (https://www.orsanco.org/data/macroinvertebrates/).

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